IOP Conference Series:
Earth and
Environmental Science

PURPOSE-LED PUBLISHING™

PAPER • **OPEN ACCESS**

Food waste management regulation in Indonesia to achieve sustainable development goals

To cite this article: F A Cahyani *et al* 2022 *IOP Conf. Ser.: Earth Environ. Sci.* **978** 012022

View the article online for updates and enhancements.

You may also like

- Eating healthy or wasting less? Reducing resource footprints of food consumption
Hanna Helander, Martin Bruckner, Sina Leipold et al.

- Conceptual framework to integrate food waste research and food systems research
Heike B Rolker, Laura Cardenas, Christian Reynolds et al.

- Are interventions for environmentally sustainable dietary behaviours effective? A review
Y Ran, U M Persson, T Lindahl et al.

This content was downloaded from IP address 202.46.151.130 on 14/03/2026 at 12:31

Food waste management regulation in Indonesia to achieve sustainable development goals

F A Cahyani [1], P Wulandari[2] and N A Putri[2]
[1] Faculty of Law University of Sultan Ageng Tirtayasa, Banten, Indonesia
[2] Faculty of Agriculture University of Sultan Ageng Tirtayasa, Banten, Indonesia

E-mail: puji.wulandari@untirta.ac.id

Abstract. This article discussed various regulation of food waste management in Indonesia. The purpose of this study was to find out the regulations governing food waste management in Indonesia. By using a normative research method that makes laws and regulations the main study material in this article. Population growth and changes in consumption patterns result in an increase of the volume, type, and characteristics of waste in Indonesia. Food waste was the biggest contributor to the waste production in Indonesia. Food waste occupied the first position in the composition of waste in Indonesia amount 40.3%. Article 28 H paragraph (1) of the 1945 Constitution of the Republic of Indonesia had mentioned that a good and healthy environment was the right of every citizen. The right to a good and healthy environment was aligned with other human rights so that their position was equal. By implementing food waste management from production to after consumption through the right strategy, it will be able to reduce the environmental impact of food waste and contribute to sustainable development goals. The results found by the authors that there were no specific rules governing food waste management in Indonesia. There were regulations governing both organic and non-organic waste in the laws that owned by Indonesia, but none had specifically discussed about food waste.

Keywords: Food waste, waste management, food manufacture, sustainable development

1. Introduction

Food is the main basic human need. Its fulfillment is part of human rights protected by the 1945 Constitution of the Republic of Indonesia. Law Number 18 of 2012 concerning Food states that food is anything that comes from biological sources of agricultural, plantation, forestry, fishery, livestock, waters, and water, both processed and unprocessed intended as food or drink for human consumption, including food additives, food raw materials, and other materials used in the process of preparing, processing, and/or making food or beverages.

World food demand is expected to increase 50-100% by 2050 in line with world population growth [1]. By 2050, the world's population is estimated to reach 9.7 billion [2]. In 2020, the Food and Agriculture Organization (FAO) stated that 720 to 811 million people are hungry. This condition also affected by the Covid-19 pandemic that has hit the whole world [3]. Meanwhile, in Indonesia, based on population census data in 2020, Indonesia's population reached 270.20 million people. This number has increased by 32.56 million people compared to the population census in 2010 [4].

Content from this work may be used under the terms of the Creative Commons Attribution 3.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

Approximately, a third of the annual food production for human consumption is either wasted or lost during the food supply chain (equivalent to 1.3 billion tons). Food loss often occurs in the production level or manufacturing level, whereas food waste is at the consumer and retail level [5]. According to FAO, food waste is the decrease of the quantity or quality of food resulting from decisions and actions by retailers, food services and consumers. While food loss is the decrease in the quantity or quality of food resulting from decisions and actions by food suppliers in the chain, excluding retail, food service providers and consumers. In developed countries, the largest contributor of food waste is at the customer stage, while in developing countries it is at the agricultural and manufacturing stages [5]. According to Directorate of Waste Management, Ministry of Environment and Forestry Republic of Indonesia, food waste occupies the first position in the composition of total waste in Indonesia (amount 40.3%), followed by plastics (17,1%), woods (14,1%), paper (11,9%), etc [6].

Food waste is a problem faced by the world today. Food waste that burdens the waste management system makes it a major contributor to the crisis that is happening on earth. The crises are climate change, loss of biodiversity, pollution, and waste [7]. Food waste does not only come from household consumption, but also comes from retail and food service. There is still a lot of food waste that is not managed properly and ends up in landfills, sewers, and so on. Apart from food, a good and healthy environment is also a part of human rights protected by the constitution. If food waste management is carried out with the right strategy, it will also affect the fulfillment of a good and healthy environment.

Sustainable production and consumption practices are needed to maintain balance between global economic growth and declining natural resources. The United Nations has developed an agenda to stimulate action through 2030 across 17 Sustainable Development Goals (SDGs). United Nation SDGs 12th urges responsible consumption and production. Food loss index and food waste index are indicators of this goal. The Food Loss Index (FLI) focuses on food losses that occur from production up to (and not including) the retail level. It measures the changes in percentage losses for a basket of 10 main commodities by country in comparison with a base period. The Food Waste Index (FWI), measures food waste at retail and consumer level (households and food service). The target of 12th SDG is by 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.

According to the Ministry of National Development Planning, food loss and waste in Indonesia are dominated by grains, namely rice, corn, wheat and related products, while the most inefficiently processed food types are vegetables, where its loss is 62.8 percent of the total domestic supply of vegetables in Indonesia. Whereas emissions from food loss and waste are equivalent to 7.29% of the average Indonesian Greenhouse Gases [8]. These data can be used as the basis for policy makers to implementing RPJMN 2020-2024 and realizing Indonesia's commitment to the Sustainable Development Goals (SDGs) [9]. Therefore, the regulation about waste management is urgently needed. The approaches to food waste management require technical and policy-based commitments, financial investments, public support and participation. Government regulation is one of the environmental factor that affecting the adoption technology chain in food waste management [9]. Political commitment is also an important factor to develop sustainable food waste management systems [10]. The aims of this study was to discuss various regulation about food waste management in Indonesia.

2. Research Method

The research method used in this study was normative research or legal research. The normative or legal research is study of documents that utilizes a qualitative method in analyzing data, and using secondary data as the sources, such as legal theories, regulations, court decisions, and books. The literature study used in this study was to analyze the laws and regulations in accordance with the existing laws and regulations in Indonesia. Data analysis method used in this tudy was qualiative method.

3. Result and Discussion

3.1. Food Waste Management

Sustainable food management is a systematic approach that seeks to reduce wasted food and its associated impacts over the entire life cycle, starting with the use of natural resources, manufacturing, sales, and consumption and ending with decisions on recovery or final disposal. Many countries has build their food management system especially in many countries in Europe and Asia. The policies and regulations for food waste management in Asia have been shaped in 4 major factor: awareness of the food waste management problem, (ii) commitment to national and international development goals, (iii) the socio-cultural and economic constraints, and (iv) the recognition of the potency to recover energy and nutrients from food [11]. Waste management worldwide is moved toward waste minimization practices through the concept of the waste management hierarchy as the effort to implement sustainability. In the waste management hierarchy, food waste prevention has the highest priority, followed by redistribution for human consumption, reuse, recycle, and energy/material recovery, and the lowest priority is disposal, see Figure 1 [12]. Different countries have adopted different strategies to address them challenge of food waste management. Technological factors and environmental factors were the most factors affecting the adoption of technologies chain in food waste management.

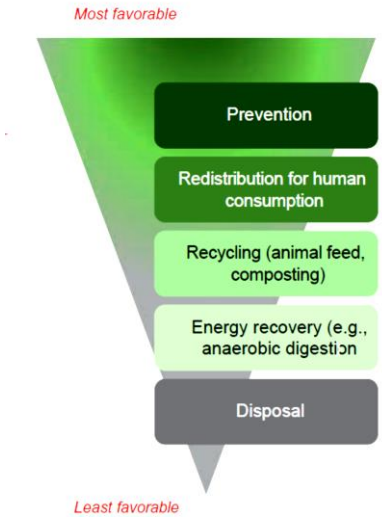

Figure 1. Food waste management hierarchy [12]

The principles used as the basis for food administration were sovereignty, independence, resilience, security, benefits, equity, sustainability and justice. The principle of sustainability means that the implementation of food is carried out consistently and sustainably by utilizing natural resources which ensures an increase in the welfare of the community for the present and the future. This means that food, both in supply and in waste management, must pay attention to the principle of sustainability, which in addition to ensuring the availability of food today, also ensures its availability in the future. With regard to food waste management, its management must also meet three important aspects of the SDGs, namely economic, social, and environmental aspects.

3.2 Food Waste Management Regulation in Indonesia

3.2.1. The 1945 Constitution of the Republic of Indonesia

In Indonesia, the right to food is one of the human rights regulated in the constitution. These human rights must be protected in order to create a prosperous country. The aims of people's welfare are to reduce poverty, improve health, and reduce unemployment [13]. In the 1945 Constitution, there is no article that mentions the right to food. However, the right to food is implicit in Article 28C paragraph (1) and 28I paragraph (4)[14].

Article 28C Paragraph (1) of the 1945 Constitution reads: "Everyone has the right to develop themselves through fulfilling basic needs, has the right to obtain education and acquire knowledge from science and technology, art and culture, in order to improve the quality of his life. and for the welfare of mankind." Article 28I Paragraph (4) of the 1945 Constitution reads: "The protection, promotion, enforcement and fulfillment of human rights are the responsibility of the state, especially the government".

Another right in the 1945 Constitution related to food waste management is the right to a good and healthy environment as stated in Article 28H paragraph (1). The article reads "Everyone has the right to live in physical and spiritual prosperity, to live, and to have a good and healthy living environment and have the right to obtain health services". Food waste management is needed to fulfill the right to a good and healthy environment. Article 28I paragraph (4) is the basis for the protection of human rights, including the right to food and the right to a good and healthy environment. Therefore, in carrying out this matter, the government is obliged to pay attention to this article on the implementation of human rights.

3.2.2. The Law Number 32 of 2009 concerning Environmental Protection and Management

Nowadays, the quality of the environment is declining. Irresponsible use of natural resources will result in threats to biodiversity, climate change, including threats to food availability. The food consumed by humans currently produces waste, the amount of which is the number one waste generation in Indonesia, which is as much as 40.3% of the total waste generation in Indonesia.

In this Law, waste is the residue of a business and/or activity [15]. However, there is no definition of food waste. Although this law discusses the environment. Food waste also contributes to the increase in greenhouse gases through the methane gas produced. Food waste that is disposed of at the final processing site will contribute to the production of methane gas. Methane gas is one of the greenhouse gases that causes global warming [16].

In Indonesia, natural resources and population can be said to be equally abundant. As natural resources are diverse, food sources will also be diverse. However, in its implementation it must still pay attention to environmental sustainability because Indonesia is in a position that is vulnerable to climate change. Climate change will have an impact on various things, one of which is a decrease in food production [17]. Apart from climate change factors, environmental sustainability also needs to be maintained by implementing waste management. Food waste that is dumped directly into the environment will result in various impacts.

The commitment of the Indonesian government to reduce the impact of climate change is marked by the issuance of Law No. 17 of 2004 concerning the Ratification of the Kyoto Protocol to the United Nations Framework Convention on Climate Change. As previously explained, methane gas from decaying waste is a contributor to the greenhouse gas effect that affects climate change. Indonesia's geographical position is vulnerable to climate change. This climate change has an impact on the disruption of water availability, the spread of pests and plant and human diseases, rising sea levels, sinking of small islands, the extinction of biodiversity, and of course the decline in food production.

3.2.3. The Law Number 18 of 2008 concerning Waste Management

The definition of food waste has not been found in the Law on Waste Management. However, the law recognizes that until now waste management is not in accordance with waste management methods and techniques with environmental insight, causing negative impacts on public health and the environment.

Waste has become a national problem so that its management must be carried out in a comprehensive and integrated manner. The management must be carried out from upstream to downstream [18] as well as food waste management. Such management must be carried out in order to provide economic, health and environmental security benefits.

3.2.4. The Presidential Regulation of the Republic of Indonesia Number 97 of 2017 concerning National Policies and Strategies on Management of Household Waste and Household-Type Waste

Article 1 of the Presidential Regulation of the Republic of Indonesia Number 97 of 2017 concerning National Policies and Strategies on Management of Household Waste and Household-Type Wastestates that household waste is waste that comes from daily activities in the household which does not include feces and specific waste. Food waste is included in the category of household waste. However, the regulation regarding food waste is not specifically regulated in this regulation.

Food waste generated at the level of distribution and consumption [19]. At the distribution level, food waste can come from markets or supermarkets. Usually, the types of vegetables and fruits that have a shape, color, or quality that do not meet the standard will be discarded. In fact, it can still be consumed. At the level of consumption, food waste is produced from leftover cooked ingredients, leftovers from food that has been served, as well as spoiled or expired food due to buying too much food.

3.2.5. Regulation of the Minister of Environment and Forestry Number P.75/MENLHK/SETJEN/KUM.1/10/2019 concerning Roadmap for Waste Reduction by Producers

Actually there is accompanying waste from food waste which also needs to be managed sustainably. The existence of food packaging cannot be underestimated because if food packaging, especially those that cannot be decomposed, are disposed of irresponsibly without being managed first, it will be bad for the environment. Currently, plastic materials for food packaging are still very popular in Indonesia, especially when the Covid-19 pandemic hit, the amount of plastic waste generated also increased. This is inseparable from the recommendation to work from home and school from home. Ordering food online has become one of the habits that has increased during the pandemic. Plastics are chosen as food and beverage packaging in delivery services because they are considered solid, spill-resistant, and inexpensive. Not only as a plastic wrapper, but also a bag to bring food to customers. There are still few food vendors who provide environmentally friendly bags for buyers. If any, the majority of the packaging bags are in the form of bags made of sponge bond which still contain 62% plastic (polypropylene) which is not biodegradable or can be decomposed in the soil. Plastic as food packaging materials, its existence has begun to be limited by the Indonesian government through the Regulation of the Minister of Environment and Forestry Number P.75/MENLHK/SETJEN/KUM.1/10/2019 concerning Roadmap for Waste Reduction by Producers.

In Article 4 paragraph (1) of the Regulation of the Minister of Environment and Forestry Number P.75/MENLHK/SETJEN/KUM.1/10/2019 concerning Roadmap for Waste Reduction by Manufacturers, it is regulated regarding waste reduction but is limited to products and product packaging that are difficult decomposed by nature. The regulation has not discussed food waste that has the potential to be produced by producers. In fact, producers included in the regulation (can be seen in Table 1) also have responsibilities in managing their food waste.

Table 1. Producers Producer in Roadmap for Waste Reduction

No.	Producer	Sector
1.	Manufacture	a. Food and beverage industry
		b. Consumer goods
		c. Cosmetics and body care industry
2.	Food and beverage service	a. Café
		b. Restaurant
		c. Catering services
		d. Hotel
3.	Retail	a. Shopping center
		b. Modern shop
		c. Public market

3.2.6. The National Mid-Term Development Plan 2020-2024 (RPJMN 2020-2024)

Food in the National Medium-Term Development Plan 2020-2024 is part of seven development agendas, namely the first agenda of "strengthening economic resilience for quality and equitable growth". In fulfilling the agenda, it is fulfilled with the management of economic resources which includes the fulfillment of food and agriculture. This increase in the management and production of food sources has a positive impact on improving the quality of community consumption and nutrition, as indicated by the Food Expectation Pattern (PPH) score of 91.3/100 (AKE, 2000 kcal/capita/day), and the declining food insecurity rate to 7.9 percent [20]. However, as the population increases, the need for food also increases. The availability of food which is influenced by seasons results in an increase in food prices. Seasons that are affected by climate change currently affect food availability. Food waste that is directly dumped into environmental media will also have an impact on climate change. So that it is not only food availability that is a priority in the RPJMN, but also waste management.

Environmental protection is a priority listed in the 2020-2024 National Mid-Term Development Plan. Efforts to build the environment are listed in Chapter VII section on Building the Environment, Improving Disaster Resilience, and Climate Change.

In Law Number 18 of 2012 concerning Food, there are principles used as the basis for food administration, namely sovereignty, independence, resilience, security, benefits, equity, sustainability and justice. The principle of sustainability means that the implementation of food is carried out consistently and sustainably by utilizing natural resources which ensures an increase in the welfare of the community for the present and the future. This means that food, both in supply and in waste management, must pay attention to the principle of sustainability, which in addition to ensuring the availability of food today, also ensures its availability in the future. With regard to food waste management, its management must also meet three important aspects of the SDGs, namely economic, social, and environmental aspects.

According to the World Food Programme, 135 million suffer from acute hunger largely due to man-made conflicts, climate change and economic downturn. The COVID-19 pandemic could now double that number, putting an additional 130 million people at risk of suffering from acute hunger by the end of 2020 [21]. The economic, social and environmental aspects that are at the heart of the SDGs must be realized through responsible and sustainable food waste management. Food waste occupies the first position in waste generation in Indonesia, which is as much as 40.3% [22]. In the National Policy and Strategy for the Management of Household Waste and Similar Household Waste as stated in the Presidential Regulation of the Republic of Indonesia Number 97 of 2017, the target for reducing household waste is 30% by 2025, while the target for handling it is 70%. Thus, food waste, which is the top waste generation in Indonesia, must be taken seriously. Food wastage is also one of the factors that threaten biodiversity, which indirectly increases the need for new agricultural land [23].

Applying the habit and mindset that food consumption must be done wisely can be the main strategy to reduce the amount of food waste that we produce. By having the mindset that choosing the right food, in this case it is low in calories but high in nutrients, can make us determine priorities based on needs. If you already have that mindset, then in providing, managing, and consuming the meaning that we choose is adjusted to our needs. This means that there is no longer any thought of buying food in excess, processing it not based on needs, and ending up as food waste.

The next strategy is a sustainable environment-oriented consumption pattern. Biodiversity, especially that of Indonesia, is indeed of various types so that food needs are also diverse, however, wise use must still be made. Natural resources, both renewable and non-renewable, must be managed sustainably. Likewise with the existing food potential, because if the management and utilization do not pay attention to the principles of a sustainable environment, the availability of food in the future will be affected. In the principle of sustainable development as shown in the picture below, of course the economic, social and environmental aspects must be balanced so that future generations can still utilize and manage what the present generation feels.

Figure 2. Sustainable Development Goals

The next strategy to reduce food waste is to utilize the potential of local food in each region. One example is sago, which used to be the main staple food in Indonesia. With Indonesia's geographical and climatic conditions, sago can grow and support environmental sustainability in Indonesia [24]. Even though there is no legal basis for food waste management, it does not mean that we are negligent. Food waste's legal home is waste law [25], so that it can refer to the Waste Management Law, the Roadmap for Waste Reduction by Producers, as well as Law Number 32 of 2009 concerning Environmental Protection and Management. By implementing the strategies previously mentioned, we can play a role in reducing the generation of food waste in Indonesia. In addition, we can also contribute to realizing the second goal of the SDGs, namely zero hunger which will also support the goals of other SDGs points such as the third, eleventh, twelfth, thirteenth, and fifteenth goals (see Figure 2).

4. Conclusion

The increase in population is in line with the increase in food waste production so that it can be used as a basis for government of Indonesia considerations to formulate regulations on food waste management in more detail so that all people can follow and play an active role in realizing the SDGs. Indonesia can adopt food waste management system from the other country while still paying attention to local wisdom and environment. Food waste management by managing food as needed, it is hoped that it will be a way that can support food security and environmental sustainability.

The strategy that can be done is to change the mindset in managing food and its waste. In addition, it also manages waste from food such as packaging that can be recycled or reused. Sustainable management by paying attention to social, economic and environmental aspects will contribute to the achievement of the Sustainable Development Goals, especially in the goal of zero hunger which will support other goals. Food waste management in Indonesia still has to be implemented based on the Law on Waste Management, although it does not specify how to manage food waste. Food waste management should also be carried out taking into account the relationship with climate change carried out based on the ratification of the Kyoto Protocol [26].

Appendices

1. The 1945 Constitution of the Republic of Indonesia
2. The Law Number 18 of 2008 concerning Waste Management
3. The Law Number 32 of 2009 concerning Environmental Protection and Management
4. The Presidential Regulation of the Republic of Indonesia Number 97 of 2017 concerning National Policies and Strategies on Management of Household Waste and Household-Type Waste
5. The Presidential Regulation of the Republic of Indonesia Number 97 of 2017 concerning National Policies and Strategies on Management of Household Waste and Household-Type Waste

6. The Presidential Regulation Number 18 of 2020 concerning National Medium-Term Development Plan 2020-2024
7. Regulation of the Minister of Environment and Forestry Number P.75/MENLHK/SETJEN/KUM.1/10/2019 concerning Roadmap for Waste Reduction by Producers

References

[1] World Economic Forum, 2015, Food: How Much Does The World Need?, https://www.weforum.org/agenda/2015/05/food-how-much-does-the-world-need/ accessed 8 November 2021
[2] United Nations, Department of Economic and Social Affairs, Population Division (2019). World Population Prospects 2019: Highlights. ST/ESA/SER.A/423.
[3] Food and Agriculture Organization, 2021, The World is at a Critical Juncture, https://www.fao.org/state-of-food-security-nutrition accessed 5 November 2021
[4] Badan Pusat Statistik Indonesia, 2020, Hasil Sensus Penduduk 2020, https://www.bps.go.id/website/images/Hasil-SP2020-ind.jpg accessed 5 November 2021
[5] Stancu, V., Haugaard, P. and Lähteenmäki, L. (2016) "Determinants of consumer food waste behaviour: Two routes to food waste). Appetite. 96:7– 17. DOI: 10.1016/j.appet.2015.08.025.
[6] Direktorat Pengelolaan Sampah. Kementerian Lingkungan Hidup dan Kehutanan. https://sipsn.menlhk.go.id/sipsn/
[7] United Nations Environment Programme (2021). Food Waste Index Report 2021. Nairobi.
[8] The Ministry of National Development Planning. http://greengrowth.bappenas.go.id/en/sustainable-food-waste-management-contributes-to-low-carbon-development-in-indonesia/
[9] Joubert, R., and Jokonya, O. 2021. A systematic literature review of factors affecting the adoption of technologies in food waste management. Procedia Computer Science 181 : 1034–1040
[10] D. Wilson, A. Scheinberg, What is good practice in solid waste management? WasteManag. Res. 28 (2010) 1055–1056.
[11] Joshi, Prabhat and Visavanatan, Chettiyappan. 2019. Sustainable management practices of food waste in Asia: Technological and policy drivers. Journal of Environmental Management 247 : 538–550
[12] Pandey, Ashok (ed). 2021. Current Developments in Biotechnology and Bioengineering. Elsevier. New York
[13] V. Hadiyono. 2020. Indonesia dalam Menjawab Konsep Negara Welfare State dan Tantangannya, Jurnal Hukum Politik dan Kekuasaan, Volume 1 No. 1, Agustus 2020
[14] Putra Perdana Ahmad Saifulloh, Gagasan Konstitusi Pangan: Urgensi Pengaturan Hak Atas Pangan Warga Negara Dalam Amandemen Kelima UUD 1945, Jurnal HAM, Volume 12 No. 2, Agustus 2021
[15] The Law Number 32 of 2009 concerning Environmental Protection and Management
[16] Envihsa, Fakultas Kesehatan Masyarakat Universitas Indonesia, 2020, Sampah dan Hubungannya Terhadap Emisi Gas RumahKaca, https://envihsa.fkm.ui.ac.id/2020/02/28/ehi-feb-march/, accessed 12 November 2021
[17] Yayasan Lembaga Konsumen Indonesia, Laporan Akhir, Kajian Kerangka Pengaturan dan Pengelolaan Makanan Berlebih di Hotel, Restaurant, dan Catering,
[18] The Law Number Number 18 of 2008 concerning Waste Management
[19] CIMSA Universitas Indonesia, Food Waste dan Pengaruhnya Terhadap Lingkungan, https://cimsa.ui.ac.id/2020/11/09/food-waste-dan-pengaruhnya-terhadap-lingkungan/, accessed 12 November 2021

[20] The Presidential Regulation Number 18 of 2020 concerning National Medium-Term Development Plan 2020-2024
[21] United Nations, Goal 2: Zero Hunger, https://www.un.org/sustainabledevelopment/hunger/, accessed 5 November 2021
[22] Sistem Informasi Pengelolaan Sampah Nasional (SIPSN), Grafik Komposisi Sampah, https://sipsn.menlhk.go.id/sipsn/, accessed 12 November 2021
[23] WWF Indonesia, Changing the Food Future by Making Wise Choice Today, 2021, https://www.wwf.id/publikasi/mengubah-masa-depan-pangan-dengan-membuat-pilihan-bijak-hari-ini-1, accessed 12 November 2021
[24] Natasa Kilibarda, Filip Djokovic, Radmila Suzic, "Food waste management — reducing and managing food waste in hospitality", Article in Meat Technology, December 2019
[25] Carrie Bradshaw, Waste Law and the Value of Food, Journal of Environmental Law, Volume 30, Issue 2, July 2018, Pages 311–331, https://doi.org/10.1093/jel/eqy009
[26] The Law Number Number 17 of 2004 concerning Kyoto Protocol to the United Nations Framework Convention on Climate Change